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Interview Assignment

SQL DATA ANALYST INTERVIEW ASSESSMENT

The assignment is a simulation of a Data Analyst interview beginning with basic SQL and progressively moving to advanced levels.

Company: Shopee

Position: Junior Data Analyst

You have been shortlisted for a Junior Data Analyst position at Shopee, an e-commerce company operating across India.

As part of the interview process, the analytics team has given you a small sample of their operational database.

The interview consists of several stages.

You are expected to demonstrate that you can:

- retrieve data using SQL
- filter and sort data
- aggregate business information
- combine tables using joins
- rank records
- calculate running totals
- understand window functions
- identify data-quality problems
- choose the appropriate SQL technique for a business question

Your first task is to run this SQL script to create the database schema in your preferred relational database software

In [ ]:

```sql
-- SHOPSPHERE
-- MySQL Database Setup

-- CREATE DATABASE
DROP DATABASE IF EXISTS shopsphere;
CREATE DATABASE shopsphere;

USE shopsphere;


-- 1. CUSTOMERS TABLE
DROP TABLE IF EXISTS customers;
CREATE TABLE customers (
    customer_id INT PRIMARY KEY,
    customer_name VARCHAR(100),
    city VARCHAR(50),
    signup_date DATE
);

-- Insert customer data
INSERT INTO customers
(customer_id, customer_name, city, signup_date)
VALUES
(101, 'Aamir Khan', 'Srinagar', '2024-01-10'),
(102, 'Sara Ali', 'Anantnag', '2024-01-15'),
(103, 'Rahul Sharma', 'Delhi', '2024-02-01'),
(104, 'Priya Singh', 'Mumbai', '2024-02-05'),
(105, 'Zainab Bhat', 'Srinagar', '2024-02-20'),
(106, 'Arjun Mehta', 'Jammu', '2024-03-01'),
(107, 'Neha Kapoor', 'Delhi', '2024-03-10'),
(108, 'Danish Mir', 'Srinagar', '2024-03-15');


-- 2. PRODUCTS TABLE
DROP TABLE IF EXISTS products;
CREATE TABLE products (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100),
    category VARCHAR(50),
    price DECIMAL(10,2)
);

-- Insert product data
INSERT INTO products
(product_id, product_name, category, price)
VALUES
(201, 'Laptop', 'Electronics', 70000),
(202, 'Smartphone', 'Electronics', 40000),
(203, 'Headphones', 'Electronics', 5000),
(204, 'Office Chair', 'Furniture', 12000),
(205, 'Desk', 'Furniture', 18000),
(206, 'Running Shoes', 'Sports', 6000),
(207, 'Football', 'Sports', 2000),
(208, 'Cricket Bat', 'Sports', 4000);
```

```sql
-- 3. ORDERS TABLE
DROP TABLE IF EXISTS orders;
CREATE TABLE orders (
    order_id INT PRIMARY KEY,
    customer_id INT,
    product_id INT,
    order_date DATE,
    quantity INT,
    amount DECIMAL(10,2),
    status VARCHAR(20),
    -- Relationships with other tables
    FOREIGN KEY (customer_id)
        REFERENCES customers(customer_id),

    FOREIGN KEY (product_id)
        REFERENCES products(product_id)
);

-- Insert order data
INSERT INTO orders
(order_id, customer_id, product_id, order_date, quantity, amount, status)
VALUES
(1001, 101, 201, '2024-03-01', 1, 70000, 'Completed'),
(1002, 102, 203, '2024-03-02', 2, 10000, 'Completed'),
(1003, 101, 203, '2024-03-05', 1, 5000, 'Completed'),
(1004, 103, 202, '2024-03-06', 1, 40000, 'Completed'),
(1005, 104, 204, '2024-03-07', 1, 12000, 'Completed'),
(1006, 105, 206, '2024-03-08', 2, 12000, 'Completed'),
(1007, 101, 205, '2024-03-10', 1, 18000, 'Completed'),
(1008, 102, 207, '2024-03-11', 2, 4000, 'Returned'),
(1009, 103, 203, '2024-03-12', 2, 10000, 'Completed'),
(1010, 104, 205, '2024-03-15', 1, 18000, 'Completed'),
(1011, 105, 207, '2024-03-16', 3, 6000, 'Completed'),
(1012, 106, 208, '2024-03-17', 1, 4000, 'Completed'),
(1013, 101, 202, '2024-03-20', 1, 40000, 'Completed'),
(1014, 107, 201, '2024-03-21', 1, 70000, 'Completed'),
(1015, 108, 206, '2024-03-22', 1, 6000, 'Completed'),
(1016, 102, 203, '2024-03-25', 1, 5000, 'Completed'),
(1017, 103, 202, '2024-03-25', 1, 40000, 'Completed'),
(1018, 104, 204, '2024-03-26', 1, 12000, 'Completed'),
(1019, 101, 203, '2024-03-28', 2, 10000, 'Completed'),
(1020, 105, 206, '2024-03-30', 1, 6000, 'Completed');


-- 4. EMPLOYEES TABLE
DROP TABLE IF EXISTS employees;
CREATE TABLE employees (
    employee_id INT PRIMARY KEY,
    employee_name VARCHAR(100),
    department VARCHAR(50),
    manager_id INT,
    salary DECIMAL(10,2),

    -- Self-referencing foreign key
    FOREIGN KEY (manager_id)
```

```
        REFERENCES employees(employee_id)
);

-- Insert managers first because manager_id refers to another employee in the s
INSERT INTO employees
(employee_id, employee_name, department, manager_id, salary)
VALUES
(1, 'Arif Malik', 'Sales', NULL, 90000),
(4, 'Fatima Khan', 'HR', NULL, 85000),
(7, 'Rahul Verma', 'IT', NULL, 120000);


-- Insert employees who report to managers
INSERT INTO employees
(employee_id, employee_name, department, manager_id, salary)
VALUES
(2, 'Bilal Ahmed', 'Sales', 1, 60000),
(3, 'Danish Shah', 'Sales', 1, 65000),
(5, 'Imran Dar', 'HR', 4, 55000),
(6, 'Neha Gupta', 'HR', 4, 60000),
(8, 'Sameer Singh', 'IT', 7, 80000),
(9, 'Tariq Bhat', 'IT', 7, 90000);


-- 6. DAILY SALES TABLE

CREATE TABLE daily_sales (
    sale_date DATE PRIMARY KEY,
    revenue DECIMAL(10,2)
);

-- Insert daily sales data
INSERT INTO daily_sales
(sale_date, revenue)
VALUES
('2024-03-01', 70000),
('2024-03-02', 10000),
('2024-03-03', 15000),
('2024-03-04', 12000),
('2024-03-05', 5000),
('2024-03-06', 40000),
('2024-03-07', 12000),
('2024-03-08', 12000),
('2024-03-09', 8000),
('2024-03-10', 18000);

--  DISCOUNTS TABLE
CREATE TABLE discounts (
    discount INT
);

INSERT INTO discounts
(discount)
VALUES
(10),
(20),
```

```
(30);


-- Verify tables using these
-- SELECT * FROM customers;
-- SELECT * FROM products;
-- SELECT * FROM orders;
-- SELECT * FROM employees;
-- SELECT * FROM daily_sales;
```

Interview Stage 1: SQL Fundamentals

We begin with basic SQL questions.

We want to make sure you understand how to retrieve and summarize data before moving to advanced analytics.

Question 1: "Let's start simple. We want to identify our customers from Srinagar. Can you retrieve their basic information?"

Return:

- customer_id
- customer_name
- city for customers from Srinagar.

Sort them alphabetically by name.

In [ ]:
```
-- Your solution
```

Question 2: "The finance team wants to identify high-value completed orders for a promotional analysis."

Find all completed orders where the order amount is greater than ₹20,000.

Return:

- order_id
- customer_id
- order_date
- amount

Sort by amount from highest to lowest.

In [ ]:
```
-- Your solution
```

Question 3: "How much revenue has each customer generated from completed orders?"

Return:

- customer_id
- total completed order value

Name the second column:

- total_spent

In [ ]:
```
-- Your solution
```

- Interviewer follow-up

"Why do we need GROUP BY here?"

In [ ]:
```
-- Expected explanation:
-- Your solution
```

Question 4: "Now identify customers who have spent more than ₹50,000"

In [ ]:
```
-- Your solution
```

- Interviewer follow-up

"Why can't we use WHERE SUM(amount) > 50000?"

In [ ]:
```
-- Expected answer:
-- Your solution
```

Interview Stage 2: Medium Level

The interviewer now wants to know whether you can combine business data from different tables.

Question 5: "The operations team doesn't want to see customer IDs. They want customer

names in their order report."

Return:

- customer name
- order ID
- order date
- amount

In [ ]:
```
-- Your solution
```

Question 6: "Now the product team wants to know which products are being sold."

Return:

- product name
- category
- order ID
- quantity
- amount

Only include completed orders.

In [ ]:
```
-- Your solution
```

Question 7: "Are there products in our catalogue that haven't been sold yet?"

Return all products, including products with no orders

In [ ]:
```
-- Your solution
```

- Interviewer follow-up

"Why did you use a LEFT JOIN instead of an INNER JOIN?"

In [ ]:
```
-- Expected answer:

-- Your solution
```

Interview Stage 3: Medium Level

The interviewer now moves to organizational data.

Question 8: "Our employees table contains the manager's employee ID rather than their name. HR wants a report showing each employee alongside their manager."

Return:

- employee name
- employee department
- manager name
- manager salary Employees without managers should also appear.

In [ ]:
```
-- Your solution
```

- Interviewer follow-up

"Why is this called a self join?"

In [ ]:
```
-- Expected answer:
-- Your solution
```

Question 9: "HR has noticed that some employees may have salaries higher than their managers. Can you identify them?"

Return:

- employee name
- employee salary
- manager name
- manager salary

In [ ]:
```
-- Your solution
```

Question 10: Discount Management

The marketing team is considering three discount levels: discount: 10, 20, 30

"Before launching the campaign, we want to simulate the selling price of every product at every discount level."

Generate every possible: `product × discount` combination

Return:

- product name
- original price
- discount
- discounted price

In [ ]:
```
-- Your solution
```

- Interviewer follow-up

"Why does this produce so many rows?"

In [ ]:
```
-- Expected answer:
-- Your solution
```

Interview Stage 4: Advanced

Question 11: "HR wants to rank employees by salary within their department."

The highest salary should receive rank 1.

Employees with the same salary should receive the same rank.

Return:

- employee name
- department
- salary
- salary rank

In [ ]:
```
-- Your solution
```

- Interview follow-up

"What would happen if we used ROW_NUMBER() instead?"

In [ ]:
```
-- Expected answer:
-- Your solution
```

Question 12: "The product manager wants the top two products in each category based on

revenue."

First calculate revenue for each product using: `quantity × product price`

Then rank products within each category

In [ ]:
```
-- Your solution
```

Question 13: "We want to understand how frequently our customers return."

For every order, show:

- customer ID
- order date
- amount
- previous order date
- next order date

In [ ]:
```
-- Your solution
```

- Interviewer asks

"What does LAG() do?"

In [ ]:
```
-- Expected answer:
-- Your solution
```

Question 14: "The finance team wants to see the company's total revenue on every daily sales record."

In [ ]:
```
-- Your solution
```

- Follow up question:

"The analyst now wants every row to contain the same grand total of revenue, instead of this running sum.

1. What should be done for that?
2. Why did we get the running sum in the first query, instead of a grand total in every row?"

In [ ]:
```
-- Your solution
```

In [ ]:
```
-- Your solution
```

Question 15: "This is the final question, and it's the one we use to distinguish candidates who understand window functions from candidates who only know the syntax."

The company wants to display the latest order amount for every customer

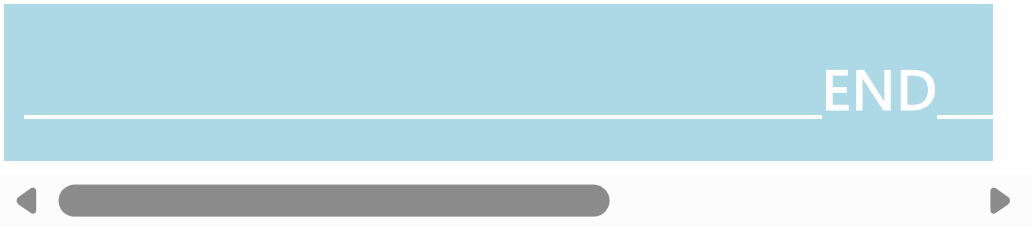
In [ ]:
```
-- Your solution
```

- Follow up question
- Is SQL is broken? Should `LAST_VALUE()` return the last value in the table?
- What is the correct query?

In [ ]:
```
-- Expected Answer
-- Your solution
```

In [ ]:
```
-- Your solution
```

Summary





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